Mandatory Assignment 2

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1 Exercise

1.1 Exercise 1.

The goal of this exercise is to construct a clean dataset for cross-sectional stock return prediction. We keep observations from January 2000 onward, select a limited set of influential predictors, document missing-value handling, and prepare a feature matrix with characteristic - macro interactions and industry dummies.

1.1.1 1a: Variable selection and sample restriction

First we read in the monthly stock characteristics data and restricting the sample to observations dated on or after January 1, 2000.

We select ten stock characteristics and four macro variables capturing key GKX themes and macroeconomic conditions, including valuation, interest rates, and credit risk.

The selected characteristics are economically interpretable and widely used in asset pricing. In particular, mom12m, mvel1, bm, and roaq capture momentum, size, value, and profitability. These are complemented by variables reflecting valuation, trading activity, risk, investment, and accounting quality. The macro variables summarize market valuation, interest rates, and credit conditions, which are standard predictors of returns.

1.1.2 1b: Summary Statistics and Missing Values

Table 1 reports summary statistics for the selected predictors.

	mean	std	min	25%	50%	75%	max
characteristic_mom12m	-0.007	0.590	-0.997	-0.526	-0.026	0.519	0.992
characteristic_mom1m	-0.004	0.604	-0.996	-0.550	-0.009	0.544	0.995
characteristic_mvel1	0.026	0.588	-1.000	-0.496	0.044	0.544	1.000
characteristic_bm	-0.065	0.531	-0.990	-0.512	-0.067	0.373	0.990

	mean	std	min	25%	50%	75%	max
characteristic_roaq	-0.014	0.560	-0.991	-0.477	-0.023	0.457	0.990
characteristic_ep	-0.011	0.558	-0.990	-0.486	-0.004	0.454	0.990
characteristic_turn	0.036	0.576	-1.000	-0.453	0.074	0.532	0.994
characteristic_beta	0.076	0.548	-0.993	-0.370	0.102	0.540	0.993
characteristic_invest	-0.006	0.533	-0.990	-0.427	-0.000	0.413	0.990
characteristic_acc	-0.008	0.545	-0.990	-0.445	-0.023	0.437	0.990
macro_dp	-4.022	0.208	-4.524	-4.107	-4.001	-3.904	-3.281
macro_ep	-3.177	0.382	-4.836	-3.317	-3.113	-2.917	-2.566
macro_tbl	0.018	0.019	0.000	0.001	0.011	0.028	0.062
macro_dfy	0.010	0.004	0.005	0.008	0.009	0.012	0.034

The selected stock characteristics are approx centered around zero and bounded close to the $[-1,1]$ interval, which is consistent with the scaling used in the provided dataset. The macro variables display much less cross-sectional variation because they are common state variables observed at the monthly level.

To document data quality, we first count missing values before imputing.

	n_missing
characteristic_mom12m	0
characteristic_mom1m	0
characteristic_mvel1	0
characteristic_bm	0
characteristic_roaq	0
characteristic_ep	0
characteristic_turn	0
characteristic_beta	0
characteristic_invest	0
characteristic_acc	0
macro_dp	0
macro_ep	0
macro_tbl	0
macro_dfy	0

Missing stock characteristics are imputed using the cross-sectional monthly median, preserving the cross section and using only contemporaneous information. Macroeconomic variables are forward- and backward-filled, reflecting their time-series nature. Observations with missing excess returns or lagged market capitalization are dropped, as these are required for later analysis.

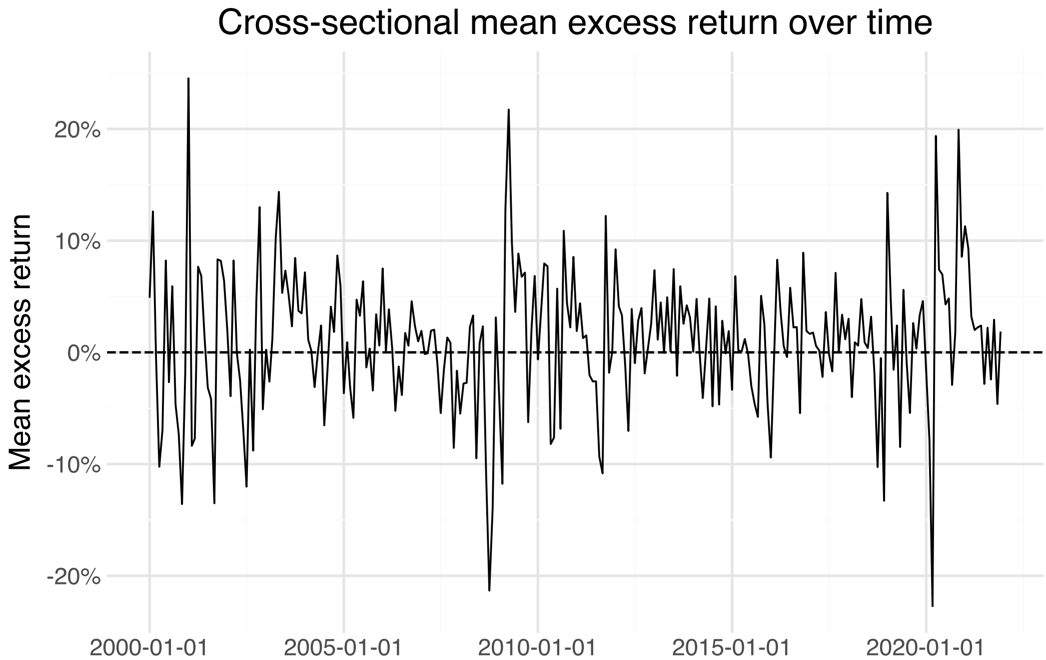
	n_missing
characteristic_mom12m	0
characteristic_mom1m	0
characteristic_mvel1	0

	n_missing
characteristic_bm	0
characteristic_roaq	0
characteristic_ep	0
characteristic_turn	0
characteristic_beta	0
characteristic_invest	0
characteristic_acc	0
macro_dp	0
macro_ep	0
macro_tbl	0
macro_dfy	0
ret_excess	0
mktcap_lag	0

Because the assignment allows the use of sic2, we recode the detailed SIC groups into the 10 major SIC divisions. This keeps the dimensionality manageable while retaining broad industry information.

	sic_division	n_obs
0	Manufacturing	442708
1	Finance	224201
2	Services	194522
3	Transportation	86802
4	Retail	64355
5	Mining	42998
6	Wholesale	31682
7	Construction	12262
8	Public Admin	9530
9	Agriculture	2545

As a simple illustration of the target variable, Figure 1 plots the cross-sectional mean excess return by month.



Overall, the sample remains large and suited for machine learning. The missing-value treatment preserves the panel structure and follows a conservative logic: firm-level predictors are imputed cross-sectionally within month, whereas macroeconomic predictors are completed along the time dimension.

1.1.3 1c: Feature matrix construction

The assignment asks for a feature matrix consisting of stock characteristics and all interactions between stock characteristics and macroeconomic variables. With 10 stock characteristics and 4 macro predictors, this yields 10+10×4=50 continuous predictors. We then add industry dummies based on the recoded SIC divisions.

	object	shape
0	X	(1111605, 59)
1	y	(1111605,)
2	meta	(1111605, 4)

	feature_group	count
0	stock characteristics	10
1	interaction terms	40
2	industry dummies	9
3	total features	59

The final feature matrix contains the 10 selected stock characteristics, 40 interaction terms between stock characteristics and macroeconomic variables, and industry dummies for the broad SIC divisions.

1.2 Exercise 2

1.2.1 Exercise 2a: Model descriptions

Elastic Net is a penalized linear regression model that combines the L_1 penalty from the Lasso and the L_2 penalty from Ridge regression. It minimizes the following objective function:

$$\min_{\beta} \frac{1}{N} \sum_{i=1}^{N} (y_i - x_i'\beta)^2 + \alpha \left(\lambda \|\beta\|_1 + (1-\lambda) \|\beta\|_2^2 \right)$$

where the first term is the mean squared prediction error and the second term is a penalty on coefficient size. This helps control overfitting in high-dimensional settings with correlated predictors. This is particularly relevant in our setting, as the feature matrix includes stock characteristics, interaction terms, and industry dummies.

The key hyperparameters are the overall penalty strength (`alpha`) and the mixing parameter (`l1_ratio`), where `l1_ratio = 1` corresponds to the Lasso and `l1_ratio = 0` corresponds to Ridge.

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Neural Network (MLP) approximates $g(z_{i,t})$ as a composition of linear transformations and nonlinear activation functions across multiple hidden layers. The model is trained by minimizing the mean squared prediction error between predicted and realized excess returns. Each neuron computes $x_k^l = f\left(\theta_0^k + \sum_j z_j \theta_j^k\right)$, where f is the ReLU activation function. The network is trained via backpropagation. The key hyperparameters are the number and size of hidden layers (`hidden_layer_sizes`), the L_2 regularization strength (`alpha`), and the maximum number of iterations (`max_iter`).

Together, the two models provide a comparison between a structured, regularized linear benchmark and a more flexible nonlinear model.

1.2.2 Exercise 2b: Time-respecting data split

```
Train:      2000-01-01 to 2013-02-01 (748,902 obs)
Validation: 2013-03-01 to 2017-07-01 (194,460 obs)
Test:       2017-08-01 to 2021-12-01 (168,243 obs)
```

```
Number of train months: 158
Number of validation months: 53
Number of test months: 53
```

The dataset is split chronologically to avoid look-ahead bias, ensuring that all training observations precede validation observations, which in turn precede test observations. The most recent 20% of months are reserved as a test set, providing a genuinely out-of-sample period for final model evaluation.

Among the remaining 80% of the data, we allocate 75% to training and 25% to validation, yielding an approximate 60/20/20 split overall. This choice reflects a trade-off between estimation accuracy and model selection. A relatively large training set is needed for reliable estimation of model parameters, especially for the Neural Network, which is more flexible and data-intensive. At the same time, a sufficiently large validation set is required to compare hyperparameter choices under realistic out-of-sample conditions.

The split is implemented at the monthly level rather than at the observation level to preserve the panel structure and ensure that all stocks from a given month are assigned to the same subset. Continuous predictors are standardized using the training data only, and the same transformation is applied to the validation and test sets to prevent data leakage.

1.2.3 Exercise 2c: Hyperparameter tuning

```
Elastic Net tuning results:
 alpha  l1_ratio  train_rmse  val_rmse
 0.100       0.8    0.199689  0.157523
 0.100       0.2    0.199689  0.157523
 0.001       0.8    0.199399  0.157715
 0.001       0.2    0.199213  0.157819

Best Elastic Net params: {'alpha': 0.1, 'l1_ratio': 0.8}
```

```
Neural Network tuning results:
hidden_layers  alpha  train_rmse  val_rmse
     (32, 16)  0.001    0.192807  0.158464
     (32, 16)  0.010    0.196407  0.158962

Best NN: layers=(32, 16), alpha=0.001
```

```
Final models fitted successfully.
Elastic Net    - val RMSE: 0.157523
Neural Network - val RMSE: 0.158464
```

Hyperparameters are tuned using grid search, where each candidate specification is evaluated on the validation set using root mean squared prediction error (RMSE), consistent with the MSPE-minimization objective in GKX. The Elastic Net grid varies both the penalty strength and the balance between L_1 and L_2 regularization (4 combinations), while the Neural Network grid varies the regularization strength for one chosen architecture (2 combinations).

The tuning results illustrate the bias–variance trade-off. For both models, lower training RMSE does not necessarily translate into lower validation RMSE. In particular, models with weaker regularization tend to fit the training data more closely but may perform worse out-of-sample due to overfitting. Conversely, stronger regularization increases bias but can improve generalization by reducing variance.

For the Elastic Net, the selected hyperparameters balance these effects by limiting coefficient magnitude while retaining predictive signal. For the Neural Network, the restricted grid reflects computational constraints and the large sample size, but still allows us to assess how regularization affects out-of-sample performance.

The final model for each method is chosen as the specification with the lowest validation RMSE, since the objective is to maximize out-of-sample predictive accuracy rather than in-sample fit.

1.2.4 Exercise 2d: Save models to disk

```
['model_neural_network.joblib']
```

To load the models for peer review, run:

```python
import joblib
final_en = joblib.load("model_elastic_net.joblib")
final_nn = joblib.load("model_neural_network.joblib")
```

The final fitted models are saved to disk using the `joblib` library to ensure reproducibility and facilitate peer review. Saving the trained models allows others to load the exact same fitted objects without re-running the entire estimation procedure, which can be computationally intensive given the large dataset.

To load the models for replication or further analysis, the following code can be used:

1.3 Exercise 3

1.3.1 Exercise 3a: Generate predicted excess returns

Using the fitted models, we generate predicted excess returns for each stock-month observation in both the validation and test samples. These predictions represent the model's estimate of expected returns based on firm characteristics, macroeconomic variables, and their interactions.

The predicted values serve as the ranking signal in the subsequent portfolio construction. Stocks with higher predicted excess returns are interpreted as more attractive investment opportunities and will be assigned to higher deciles, while stocks with lower predicted values will be placed in lower deciles.

1.3.2 Exercise 3b: Decile portfolio sorts

Stocks are sorted into decile portfolios each month based on their predicted excess returns. Within each month, stocks are ranked according to the model's predictions and then divided into ten groups of approximately equal size, ranging from the lowest predicted returns (decile 0) to the highest (decile 9).

To ensure stable portfolio formation, predicted values are first ranked within each month before applying quantile-based sorting. This avoids issues where many identical predictions would otherwise lead to dropped bins in the quantile procedure. Months with very few observations are excluded from the sorting step.

This procedure produces a cross-sectional ranking of stocks that reflects the model's implied expected return signal, which is used in the subsequent construction of long-short portfolios.

1.3.3 Exercise 3c: Long-short portfolio

```
Long-short portfolios constructed.
Validation period: 2013-03-01 to 2017-07-01
Test period:       2017-08-01 to 2021-12-01
Number of valid validation months (EN): 53
Number of valid test months (EN): 53
```

1.3.4 Exercise 3d: Performance metrics

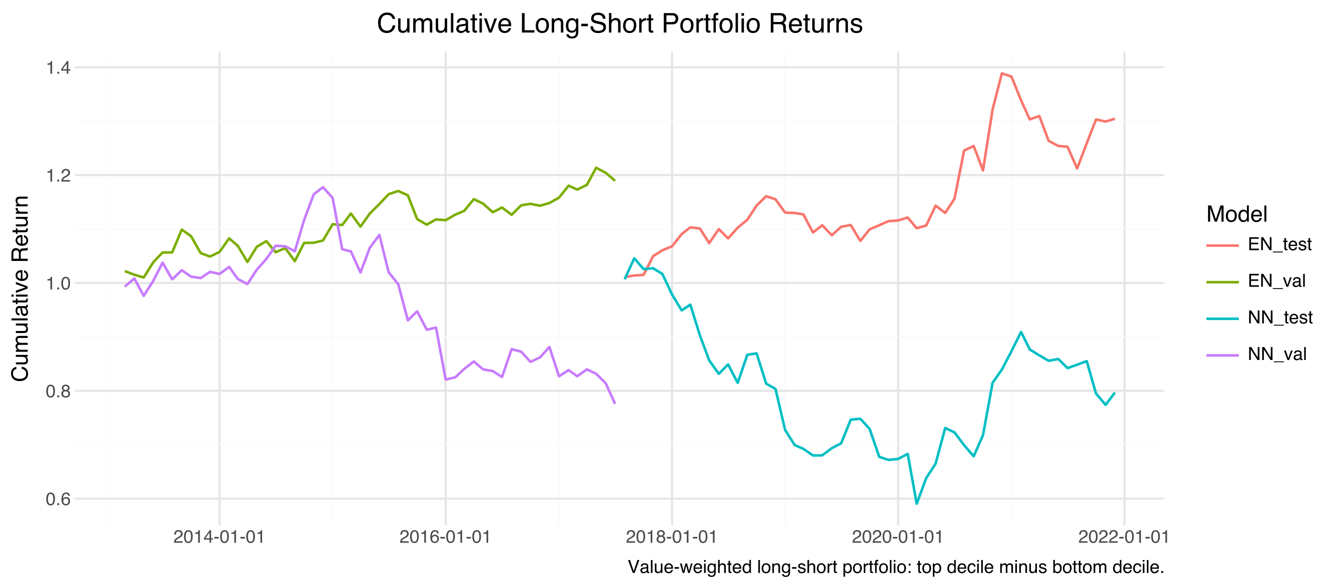
```
Elastic Net - Validation
  Annualized mean excess return: 0.0410 (4.10%)
  Annualized Sharpe ratio:       0.6976
  CAPM alpha (annualized):       0.0405 (4.05%)
  CAPM alpha t-stat:             1.3369

Neural Net  - Validation
  Annualized mean excess return: -0.0507 (-5.07%)
  Annualized Sharpe ratio:       -0.4420
  CAPM alpha (annualized):       -0.0922 (-9.22%)
  CAPM alpha t-stat:             -1.6196

Elastic Net - Test
  Annualized mean excess return: 0.0641 (6.41%)
  Annualized Sharpe ratio:       0.7228
  CAPM alpha (annualized):       0.0449 (4.49%)
  CAPM alpha t-stat:             1.0329

Neural Net  - Test
  Annualized mean excess return: -0.0382 (-3.82%)
```

```
  Annualized Sharpe ratio:       -0.2336
  CAPM alpha (annualized):       -0.0769 (-7.69%)
  CAPM alpha t-stat:             -0.9641

Market benchmark - Validation
  Annualized mean: 13.61%
  Sharpe ratio:    1.3293

Market benchmark - Test
  Annualized mean: 17.48%
  Sharpe ratio:    1.0233
```

1.3.5 Exercise 3e: Portfolio turnover

```
Monthly turnover - Elastic Net:    Val=0.007, Test=0.008
Monthly turnover - Neural Network: Val=0.341, Test=0.408
```

1.3.6 Exercise 3f: Model comparison



Both models are compared across Sharpe ratio, CAPM alpha, and portfolio turnover. The Elastic Net is the linear benchmark and tends to have lower turnover due to its smooth, stable coefficient structure. The Neural Network captures nonlinear interactions but may overfit to short-lived patterns, leading to higher turnover and potentially lower net-of-cost returns. If the Neural Network's alpha does not sufficiently exceed the Elastic Net's to compensate for higher turnover costs, the Elastic Net is preferable from an investor's perspective.

References